

Prereq: Lab 06: Jenkins Live

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Lab 01: Automated Integration Foundations



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Lab 08: Backup & Restore

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Objective

Prove disaster recovery: back up JENKINS_HOME, destroy the controller, restore on fresh infra, replay the smoke pipeline.

Prerequisites

- Lab 06 controller with job history; concepts: Jenkins Advanced backup section.

Architecture

Running controller → stop → archive jenkins_home volume → wipe → fresh controller → restore archive → verify jobs, credentials, history → run smoke pipeline.

Setup

Note one job's last build number and one credential ID — these are what you will verify after restore.

Step 1 — Back up

Execute

```
docker compose stop jenkins docker run --rm -v jenkins-lab_jenkins_home:/data -v "$PWD":/out alpine \ tar czf /out/jenkins-backup.tar.gz -C /data . docker compose start jenkins
```

Expected Result

jenkins-backup.tar.gz exists, timestamped; controller restarts normally.

Why

Offline backup = consistent snapshot; live copies risk half-written state.

Step 2 — Destroy and restore

Execute

```
docker compose down -v # destroy EVERYTHING, including the volume
docker compose up -d # fresh controller
docker compose stop jenkins
docker run --rm -v jenkins-lab_jenkins_home:/data -v "$PWD":/in alpine \
tar xzf /in/jenkins-backup.tar.gz -C /data
docker compose start jenkins
```

Expected Result

Jobs, build history, credentials, and plugins all present; recorded build number matches.

Why

Backup untested = backup imaginary. This drill is the quarterly DR practice from the advanced doc.

Verification

- [] Job list + history identical (check the recorded build number).
- [] Credential ID present and usable.
- [] Smoke pipeline (Lab 06's) runs green on the restored controller.

Failure Scenarios

- Plugin version drift: restore on a different image tag — pin `jenkins:1ts` digests for controller + backup.
- Permission errors: volume owned by wrong UID — `chown -R 1000:1000` on the volume.

Cleanup

Store one backup off-machine; delete the rest.

What This Proved

Recovery measured, not assumed. Labs complete — return to the Topic Index.

Interview Notes

- The restore proof: which pipeline you run post-restore and what green actually verifies.
- Plugin drift and UID ownership — the two restore failures and their one-line fixes.

RELATED BOOKS



Lab 01: Automated Integration Foundations

45 min · INT

→ Next in path



Lab 06: Jenkins Live

60 min · INT

→ Prerequisite



Lab 07: Shared Library

60 min · ADV

→ Previous in path



Lab 01: First Pipeline

45 min · BEG

→ Same path